Supplemental Materials for "Modeling the Calcium Sequestration System in Isolated Guinea Pig Cardiac Mitochondria"

Jason N. Bazil[1], Christoph A. Blomeyer[2], Ranjan K. Pradhan[1], Amadou K.S. Camara[2] and Ranjan K. Dash[1]

[1]Biotechnology and Bioengineering Center and Department of Physiology
[2]Department of Anesthesiology
Medical College of Wisconsin, Milwaukee, WI 53226.

To whom correspondence should be addressed: Ranjan K. Dash, Biotechnology and Bioengineering Center and Department of Physiology, 8701 Watertown Plank Rd., Milwaukee, WI, USA, Tel: (414) 955-4497; Fax: (414) 955-6568; E-mail: rdash@mcw.edu

The Supplemental Materials is split into three supplements. Supplement 1 introduces the trend line fits to the data. Supplement 2 presents a detailed description of the integrated model used to simulate the results in the main paper. Supplement 3 addresses the NCE proton dependence in the context of the Ca^{2+} uptake, sequestration and release dynamics observed in the data.

Supplement 1: Model Independent Analysis of Extra-Mitochondrial and Mitochondrial Ca^{2+} Responses to $CaCl_2$ and NaCl Bolus Additions

Figure S1 demonstrates the trend curve fits to the data and their use in extracting the rates of extra-mitochondrial and mitochondrial [Ca^{2+}] changes. The trend lines are multi-exponential functions of the form shown in Eq. 9 of the main paper. The rates were estimated by evaluating the time-derivative of the trend line functions. These rates were used to estimate the mitochondrial Ca^{2+} buffering power as described in the main paper. For Figure 3 of the main paper, the buffering power is estimated at 462 s. This time is indicated by the vertical gray line in Figure S1.

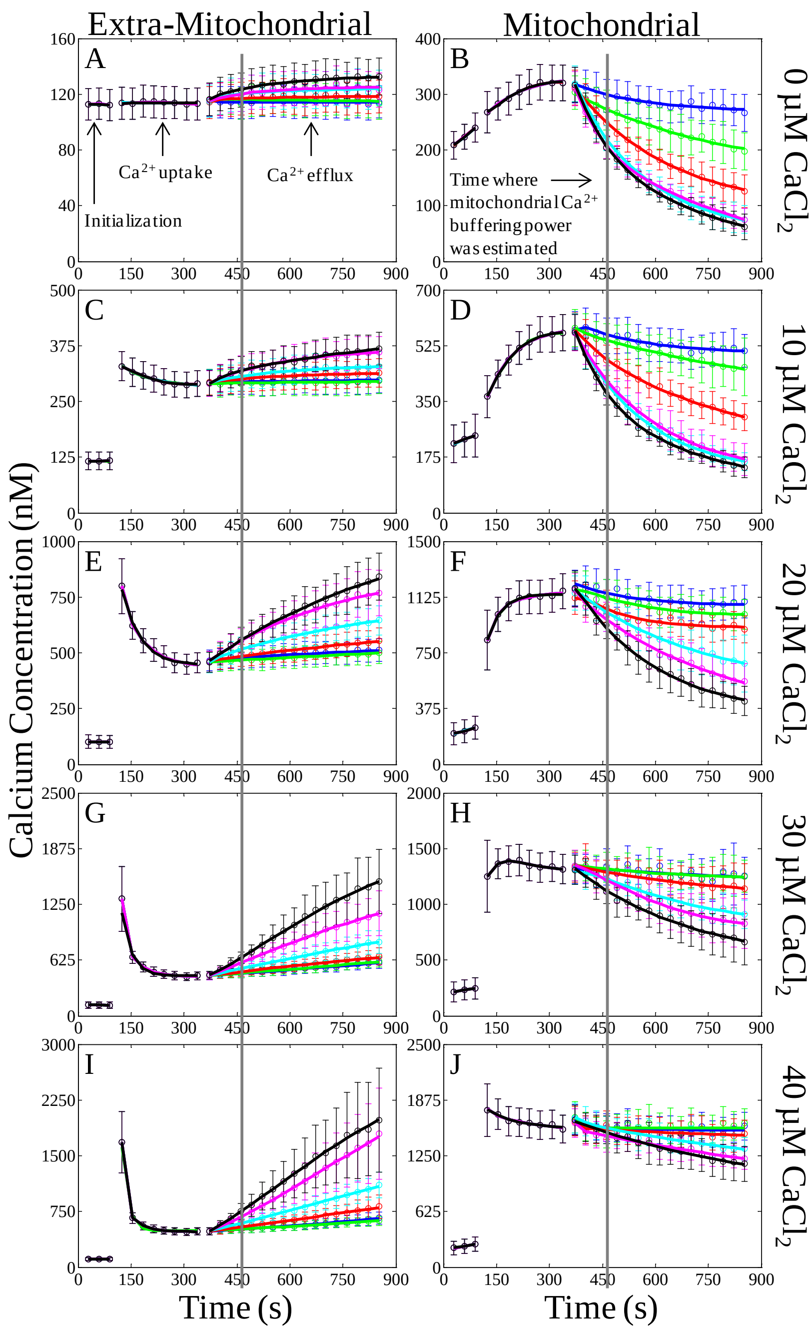

Figure S1. Trend curve fits to the data for the $CaCl_2$ and NaCl bolus additions. The trend line fits to the data for each bolus $CaCl_2$ addition are plotted by row, while the extra-mitochondrial (left) and mitochondrial (right) Ca^{2+} concentrations are plotted by column. The three separate phases (initialization, Ca^{2+} uptake and Ca^{2+} efflux) in the data sets are indicated by the text in panel A. The time where the mitochondrial Ca^{2+} buffering power was estimated (462 s) is identified by the vertical gray line and indicated by the text in panel B. The color of each line corresponds to the following NaCl bolus addition: blue, 0 mM; green 1 mM; red, 2.5 mM; cyan, 5 mM; magenta, 10 mM; black, 20 mM. Extra-mitochondrial buffer contained 40 μM EGTA.

Supplement 2: Integrated Model of Mitochondrial Bioenergetics and Cation Dynamics

The present model is extended from the previous model of mitochondrial bioenergetics and Ca^{2+} handling of Dash and Beard [1], by accounting for the binding of protons (H^+) and metal ions (Na^+, K^+, Mg^{2+} and Ca^{2+}) to metabolites (ATP, ADP and Pi) and cation buffering proteins. Several flux expressions and model parameters pertaining to oxidative phosphorylation and the Na^+-Ca^{2+} cycle were also updated based on the recent works of Wu et al. [2], Bazil and Dash [3], and Pradhan et al. [4, 5], as laid out in this model supplemental material section. This supplement consists of three parts. Part 1 lists all the state variables comprising the model, dissociation constants, and physiochemical and general model parameters. Part 2 introduces the set of 10 non-linear ODEs, four algebraic conservation expressions (for mitochondrial ATP, NADH and UQH_2 and inter-membrane space $CytC^{2+}$), and 10 non-linear cation ODEs (mitochondrial and extra-mitochondrial H^+, K^+, Mg^{2+}, Ca^{2+} and Na^+). Part 3 presents the model rate equations and the associated parameter definitions and values.

The outer-mitochondrial membrane (OMM) is highly permeable to ions, metabolites and substrates of low molecular weight. As such, all permeable inter-membrane space (IMS) state variables were replaced with their respective extra-mitochondrial counterparts, and thus not included in the model. This has the benefit of making the system of non-linear ODEs less stiff and shortens simulation time without significantly altering the simulation results. The only IMS state variables explicitly simulated in the model are oxidized and reduced forms of cytochrome c ($CytC^{3+}$ and $CytC^{2+}$).

S2.1 - Model State Variables, Dissociation Constants, and Physiochemical and General Parameter Values

The state variables used in the model, their definitions, and their units are provided in Table S2.1.1. All the dissociation constants used in the model are presented in Table S2.1.2. Most of the dissociation constants are taken from Wu et al. [2] and corrected for appropriate temperature ($T = 25\ ^oC$) and ionic strength ($I = 0.17$ M), corresponding to the present experiments. The EGTA dissociation constants are taken from [6]. The physiochemical and general model parameters are presented in Table S2.1.3.

Table S2.1.1. Model State Variables

State Variable	Definition	Units
$\Delta\Psi$	Mitochondrial membrane potential	mV
RO	Fraction of RaM in rest and open states	unitless
Mitochondrial State Variables		
$[H^+]_m$	Mitochondrial free proton concentration	M
$[K^+]_m$	Mitochondrial free potassium concentration	M
$[Na^+]_m$	Mitochondrial free sodium concentration	M
$[Mg^{2+}]_m$	Mitochondrial free magnesium concentration	M
$[Ca^{2+}]_m$	Mitochondrial free calcium concentration	M
$[ATP]_m$	Total mitochondrial ATP concentration	M
$[ADP]_m$	Total mitochondrial ADP concentration	M
$[Pi]_m$	Total mitochondrial Pi concentration	M
$[NADH]_m$	Total mitochondrial NADH concentration	M
$[NAD]_m$	Total mitochondrial NAD concentration	M
$[UQH_2]_m$	Total mitochondrial ubiquinol concentration	M
$[UQ]_m$	Total mitochondrial ubiquinone concentration	M
Intermembrane Space (IMS) State Variables		
$[CytC^{2+}]_i$	Total IMS cytochrome c^{2+} (reduced) concentration	M
$[CytC^{3+}]_i$	Total IMS cytochrome c^{3+} (oxidized) concentration	M
Extra-Mitochondrial State Variables		
$[H^+]_e$	Extra-mitochondrial free proton concentration	M
$[K^+]_e$	Extra-mitochondrial free potassium concentration	M
$[Na^+]_e$	Extra-mitochondrial free sodium concentration	M
$[Mg^{2+}]_e$	Extra-mitochondrial free magnesium concentration	M
$[Ca^{2+}]_e$	Extra-mitochondrial free calcium concentration	M
$[ATP]_e$	Total extra-mitochondrial ATP concentration	M
$[ADP]_e$	Total extra-mitochondrial ADP concentration	M
$[Pi]_e$	Total extra-mitochondrial Pi concentration	M

Table S2.1.2. Dissociation Constants at 25 ºC, I = 0.17 M

Parameter	Definition	Value	Reference
K_{ATP}^{H}	Proton ATP binding constant	$10^{-6.33}$ M	[2]
K_{ATP}^{Na}	Sodium ATP binding constant	$10^{-1.16}$ M	[2]
K_{ATP}^{K}	Potassium ATP binding constant	$10^{-1.02}$ M	[2]
K_{ATP}^{Mg}	Magnesium ATP binding constant	$10^{-3.88}$ M	[b]
K_{ATP}^{Ca}	Calcium ATP binding constant	$10^{-3.55}$ M	[b]
K_{ADP}^{H}	Proton ADP binding constant	$10^{-6.26}$ M	[2]
K_{ADP}^{Na}	Sodium ADP binding constant	$10^{-1.00}$ M	[2]
K_{ADP}^{K}	Potassium ADP binding constant	$10^{-0.89}$ M	[2]
K_{ADP}^{Mg}	Magnesium ADP binding constant	$10^{-3.00}$ M	[b]
K_{ADP}^{Ca}	Calcium ADP binding constant	$10^{-2.63}$ M	[b]
K_{Pi}^{H}	Proton Pi binding constant	$10^{-6.62}$ M	[2]
K_{Pi}^{Na}	Sodium Pi binding constant	$10^{-0.12}$ M	[2]
K_{Pi}^{K}	Calcium Pi binding constant	$10^{-0.42}$ M	[2]
K_{Pi}^{Mg}	Sodium Pi binding constant	$10^{-1.66}$ M	[b]
K_{Pi}^{Ca}	Calcium Pi binding constant	$10^{-1.45}$ M	[b]
K_{EGTA}^{H}	1st proton EGTA binding constant	$10^{-9.58}$ M	[6][a]
K_{HEGTA}^{H}	2nd proton EGTA binding constant	$10^{-8.95}$ M	[6][a]
K_{EGTA}^{Ca}	Calcium EGTA binding constant	$10^{-10.97}$ M	[6][a]
K_{MOPS}^{H}	Proton MOPS binding constant	$10^{-7.14}$ M	[b]

[a] Note that these values are for I=0.1M and T=20°C. Corrections were not made due to insufficiently enthalpy data. Binding of Na^+, K^+ and Mg^{2+} to EGTA were assumed negligible under the simulated conditions.
[b] NIST Standard Reference Database 46.

Table S2.1.3. Physiochemical and General Model Parameters

Parameter	Definition	Value	Reference
R	Ideal gas constant	$8.314\mathrm{x}10^{-3}$ kJ/K/mol	Physical constant
T	Temperature	298.15 K	Physical constant
F	Faraday's constant	$96.487\mathrm{x}10^{-3}$ kJ/mV/mol	Physical constant
z_{Na}	Sodium valence	+1	Physical constant
z_{Ca}	Calcium valence	+2	Physical constant
z_{T}	ATP valence	-4	Physical constant
z_{D}	ADP valence	-3	Physical constant
C_{mito}	IMM capacitance	0.0248 nmol/mg/mV	[2][a]
N_{tot}	Total NAD concentration	3 mM	[2]
Q_{tot}	Total UQ concentration	1.4 mM	[2]
C_{tot}	Total CytC concentration	2.7 mM	[2]
A_{tot}	Total AdN concentration	10 mM	[2]
Vol_m	Matrix H_2O volume	1 µl/mg	[b]
Vol_i	IMS H_2O volume	0.11 µl/mg	[b]
Vol_e	buffer H_2O volume	2 ml/mg	[d]
[$EGTA$]	Extra-mitochondrial EGTA concentration	40 µM	[7][d]
[$MOPS$]	Extra-mitochondrial MOPS concentration	20 mM	[7]

Abbreviations: IMM, inner mitochondrial membrane; NAD, nicotinamide adenine dinucleotide; UQ, ubiquinone; CytC, cytochrome c; AdN, adenine nucleotide.

[a] Adjusted units to nmol/mg/mV from mol/l mito/mV using $3.7\mathrm{x}10^{-6}$ l mito/mg (mg is used for mg mitochondrial protein).

[b] Approximate mitochondrial matrix and IMS volumes.

[c] Based on a mitochondrial load of 0.5 mg in a 1 ml chamber.

[d] EGTA was 10 µM for the set-point simulations.

S2.2 - Model Differential-Algebraic Equations

S2.2A - Model Differential Equations (Bioenergetics)

Mitochondrial Membrane Potential:

$$\frac{d\Delta\Psi}{dt} = \left(4J_{C1} + 2J_{C3} + 4J_{C4} - nH_{F_1F_O} J_{F_1F_O} - J_{ANT} - J_{Hleak} - 4J_{CU} - J_{RaM} - J_{NCE}\right) / C_{mito} \tag{2.1}$$

Rapid Mode of Calcium Uptake State:

$$\frac{dRO}{dt} = \left(g_{Ca} k_{I_1O} + (1 + g_{Ca}) k_{I_2R}\right)(1 - RO) - \left(f_{Ca} k_{OI_1} + (1 - f_{Ca}) k_{RI_2}\right) RO \tag{2.2}$$

Mitochondrial State Variables:

$$\frac{d[ADP]_m}{dt} = \left(J_{ANT} - J_{F_1F_O}\right) / Vol_m \tag{2.3}$$

$$\frac{d[Pi]_m}{dt} = \left(J_{PIC} - J_{F_1F_O}\right) / Vol_m \tag{2.4}$$

$$\frac{d[NAD]_m}{dt} = \left(J_{C1} - J_{DH}\right) / Vol_m \tag{2.5}$$

$$\frac{d[UQ]_m}{dt} = \left((2-1)J_{C3} - J_{C1}\right) / Vol_m \tag{2.6}$$

Inter-Membrane Space (IMS) State Variables:

$$\frac{d[CytC^{3+}]_i}{dt} = \left(2J_{C4} - 2J_{C3}\right) / Vol_i \tag{2.7}$$

Extra-Mitochondrial State Variables:

$$\frac{d[ATP]_e}{dt} = -J_{ATP} / Vol_e \tag{2.8}$$

$$\frac{d[ADP]_e}{dt} = -J_{ADP} / Vol_e \tag{2.9}$$

$$\frac{d[Pi]_e}{dt} = -J_{Pi} / Vol_e \tag{2.10}$$

S2.2B - Mitochondrial Conservation Algebraic Equations

The mitochondrial species for adenine nucleotides (AdNs), nicotinamide adenine dinucleotides (NADH), ubiquinone (UQH_2), and reduced cytochrome c ($CytC^{2+}$) are conserved in the model. The conservation is implemented by using an algebraic expression to govern the conservation. This is done using a mass matrix with the integration algorithm and setting the appropriate rows equal to a zero row vector.

$$[ATP]_m = A_{tot} - [ADP]_m \tag{2.11}$$

$$[NADH]_m = N_{tot} - [NAD]_m \tag{2.12}$$

$$[UQH_2]_m = Q_{tot} - [UQ]_m \tag{2.13}$$

$$[CytC^{2+}]_i = C_{tot} - [CytC^{3+}]_i \tag{2.14}$$

S2.2C - Model Differential Equations (Cations)

The cation differential equations for the mitochondrial and extra-mitochondrial compartments are derived using the method outlined in Vinnakota et al. [8]. The general method will be presented with the understanding that compartment specific concentrations and transport rates are to be used where appropriate. In some cases, compartment specific equations will be given. Due to the large expressions resulting from the derivation, the method used to obtain them is presented versus explicitly showing all the terms that enter the differential equations (each differential equation would take several pages). The equations used in the model are obtained by solving the linear system of equations given in Equation 2.15.

The generalized system of equations relating the cation differential equations is

$$\begin{bmatrix} \frac{d[H^+]}{dt} \\ \frac{d[Na^+]}{dt} \\ \frac{d[K^+]}{dt} \\ \frac{d[Mg^{2+}]}{dt} \\ \frac{d[Ca^{2+}]}{dt} \end{bmatrix} = \begin{bmatrix} 1+H_{\partial H} & H_{\partial K} & H_{\partial Na} & H_{\partial Mg} & H_{\partial Ca} \\ Na_{\partial H} & 1+Na_{\partial Na} & Na_{\partial K} & Na_{\partial Mg} & Na_{\partial Ca} \\ K_{\partial H} & K_{\partial Na} & 1+K_{\partial K} & K_{\partial Mg} & K_{\partial Ca} \\ Mg_{\partial H} & Mg_{\partial K} & Mg_{\partial Na} & 1+Mg_{\partial Mg} & Mg_{\partial Ca} \\ Ca_{\partial H} & Ca_{\partial K} & Ca_{\partial Na} & Ca_{\partial Mg} & 1+Ca_{\partial Ca} \end{bmatrix}^{-1} \begin{bmatrix} \Phi_H \\ \Phi_{Na} \\ \Phi_K \\ \Phi_{Mg} \\ \Phi_{Ca} \end{bmatrix}, \tag{2.15}$$

where $X_{\partial Y}$ is the partial derivative of the concentration of *bound* X with respect to Y (Y can equal X) and Φ_X is the flux of X into or out of the compartment.

Assuming higher order cation binding is negligible, the partial derivative expressions are defined below where N_r is the number of reactants, L_i is the i^{th} ligand, K_i^j is the dissociation constant for the i^{th} ligand and j^{th} cation couple and P_i is the binding polynomial for the *i*th ligand as originally defined by Alberty [9].

$$H_{\partial H} = \frac{\partial [H_{bound}]}{\partial [H^+]} = \sum_{i=1}^{Nr} \frac{[L_i]\left(1+[Na^+]/K_i^{Na} + [K^+]/K_i^K + [Mg^{2+}]/K_i^{Mg} + [Ca^{2+}]/K_i^{Ca}\right)}{K_i^H P_i^2} \tag{2.16}$$

$$H_{\partial K} = \frac{\partial [H_{bound}]}{\partial [K^+]} = -[H^+] \sum_{i=1}^{Nr} \frac{[L_i]/K_i^H}{K_i^K P_i^2} \tag{2.17}$$

$$H_{\partial Na} = \frac{\partial [H_{bound}]}{\partial [Na^+]} = -[H^+] \sum_{i=1}^{Nr} \frac{[L_i]/K_i^H}{K_i^{Na} P_i^2} \tag{2.18}$$

$$H_{\partial Mg} = \frac{\partial [H_{bound}]}{\partial [Mg^{2+}]} = -[H^+] \sum_{i=1}^{Nr} \frac{[L_i]/K_i^H}{K_i^{Mg} P_i^2} \tag{2.19}$$

$$H_{\partial Ca} = \frac{\partial [H_{bound}]}{\partial [Ca^{2+}]} = -[H^+] \sum_{i=1}^{Nr} \frac{[L_i]/K_i^H}{K_i^{Ca} P_i^2} \tag{2.20}$$

$$Na_{\partial H} = \frac{\partial [Na_{bound}]}{\partial [H^+]} = -[Na^+] \sum_{i=1}^{Nr} \frac{[L_i]/K_i^{Na}}{K_i^H P_i^2} \tag{2.21}$$

$$Na_{\partial Na} = \frac{\partial [Na_{bound}]}{\partial [Na^+]} = \sum_{i=1}^{Nr} \frac{[L_i]\left(1+[H^+]/K_i^H + [K^+]/K_i^K + [Mg^{2+}]/K_i^{Mg} + [Ca^{2+}]/K_i^{Ca}\right)}{K_i^{Na} P_i^2} \tag{2.22}$$

$$Na_{\partial K} = \frac{\partial [Na_{bound}]}{\partial [K^+]} = -[Na^+] \sum_{i=1}^{Nr} \frac{[L_i]/K_i^{Na}}{K_i^K P_i^2} \tag{2.23}$$

$$Na_{\partial Mg} = \frac{\partial [Na_{bound}]}{\partial [Mg^{2+}]} = -[Na^+] \sum_{i=1}^{Nr} \frac{[L_i]/K_i^{Na}}{K_i^{Mg} P_i^2} \tag{2.24}$$

$$Na_{\partial Ca} = \frac{\partial [Na_{bound}]}{\partial [Ca^{2+}]} = -[Na^+] \sum_{i=1}^{Nr} \frac{[L_i]/K_i^{Na}}{K_i^{Ca} P_i^2} \tag{2.25}$$

$$K_{\partial H} = \frac{\partial [K_{bound}]}{\partial [H^+]} = -[K^+] \sum_{i=1}^{Nr} \frac{[L_i]/K_i^K}{K_i^H P_i^2} \tag{2.26}$$

$$K_{\partial Na} = \frac{\partial [K_{bound}]}{\partial [Na^+]} = -[K^+] \sum_{i=1}^{Nr} \frac{[L_i]/K_i^K}{K_i^{Na} P_i^2} \tag{2.27}$$

$$K_{\partial K} = \frac{\partial [K_{bound}]}{\partial [K^+]} = \sum_{i=1}^{Nr} \frac{[L_i]\left(1+[H^+]/K_i^H + [Na^+]/K_i^{Na} + [Mg^{2+}]/K_i^{Mg} + [Ca^{2+}]/K_i^{Ca}\right)}{K_i^K P_i^2} \tag{2.28}$$

$$K_{\partial Mg} = \frac{\partial [K_{bound}]}{\partial [Mg^{2+}]} = -[K^+] \sum_{i=1}^{Nr} \frac{[L_i]/K_i^K}{K_i^{Mg} P_i^2} \tag{2.29}$$

$$K_{\partial Ca} = \frac{\partial [K_{bound}]}{\partial [Ca^{2+}]} = -[K^{+}] \sum_{i=1}^{Nr} \frac{[L_i]/K_i^{K}}{K_i^{Ca} P_i^2} \tag{2.30}$$

$$Mg_{\partial H} = \frac{\partial [Mg_{bound}]}{\partial [H^{+}]} = -[Mg^{2+}] \sum_{i=1}^{Nr} \frac{[L_i]/K_i^{Mg}}{K_i^{H} P_i^2} \tag{2.31}$$

$$Mg_{\partial Na} = \frac{\partial [Mg_{bound}]}{\partial [Na^{+}]} = -[Mg^{2+}] \sum_{i=1}^{Nr} \frac{[L_i]/K_i^{Mg}}{K_i^{Na} P_i^2} \tag{2.32}$$

$$Mg_{\partial K} = \frac{\partial [Mg_{bound}]}{\partial [K^{+}]} = -[Mg^{2+}] \sum_{i=1}^{Nr} \frac{[L_i]/K_i^{Mg}}{K_i^{K} P_i^2} \tag{2.33}$$

$$Mg_{\partial Mg} = \frac{\partial [Mg_{bound}]}{\partial [Mg^{2+}]} = \sum_{i=1}^{Nr} \frac{[L_i]\left(1 + [H^{+}]/K_i^{H} + [Na^{+}]/K_i^{Na} + [K^{+}]/K_i^{K} + [Ca^{2+}]/K_i^{Ca}\right)}{K_i^{Mg} P_i^2} \tag{2.34}$$

$$Mg_{\partial Ca} = \frac{\partial [Mg_{bound}]}{\partial [Ca^{2+}]} = -[Mg^{2+}] \sum_{i=1}^{Nr} \frac{[L_i]/K_i^{Mg}}{K_i^{Ca} P_i^2} \tag{2.35}$$

$$Ca_{\partial H} = \frac{\partial [Ca_{bound}]}{\partial [H^{+}]} = -[Ca^{2+}] \sum_{i=1}^{Nr} \frac{[L_i]/K_i^{Ca}}{K_i^{H} P_i^2} \tag{2.36}$$

$$Ca_{\partial Na} = \frac{\partial [Ca_{bound}]}{\partial [Na^{+}]} = -[Ca^{2+}] \sum_{i=1}^{Nr} \frac{[L_i]/K_i^{Ca}}{K_i^{Na} P_i^2} \tag{2.37}$$

$$Ca_{\partial K} = \frac{\partial [Ca_{bound}]}{\partial [K^{+}]} = -[Ca^{2+}] \sum_{i=1}^{Nr} \frac{[L_i]/K_i^{Ca}}{K_i^{K} P_i^2} \tag{2.38}$$

$$Ca_{\partial Mg} = \frac{\partial [Ca_{bound}]}{\partial [Mg^{2+}]} = -[Ca^{2+}] \sum_{i=1}^{Nr} \frac{[L_i]/K_i^{Ca}}{K_i^{Mg} P_i^2} \tag{2.39}$$

$$Ca_{\partial Ca} = \frac{\partial [Ca_{bound}]}{\partial [Ca^{2+}]} = \sum_{i=1}^{Nr} \frac{[L_i]\left(1 + [H^{+}]/K_i^{H} + [Na^{+}]/K_i^{Na} + [K^{+}]/K_i^{K} + [Mg^{2+}]/K_i^{Mg}\right)}{K_i^{Ca} P_i^2} \tag{2.40}$$

and

$$P_i = 1 + [H^{+}]/K_i^{H} + [Na^{+}]/K_i^{Na} + [K^{+}]/K_i^{K} + [Mg^{2+}]/K_i^{Mg} + [Ca^{2+}]/K_i^{Ca}\,. \tag{2.41}$$

Additional Buffering in the Mitochondrial Compartment

To account for additional buffering not attributed to metabolites and substrates in the mitochondrial compartment, Eqs. 2.16, 2.34 and 2.40 are modified by adding the following terms:

$$H_{\partial H} = H_{\partial H} + \frac{[B_H]}{K_H\left(1+[H^+]/K_H\right)^2} \tag{2.16x}$$

$$Mg_{\partial Mg} = Mg_{\partial Mg} + \frac{[B_{Mg}]}{K_{Mg}\left(1+[Mg^{2+}]/K_{Mg}\right)^2} \tag{2.31x}$$

$$Ca_{\partial Ca} = Ca_{\partial Ca} + \sum_i \frac{n_i^2[Ca^{2+}]^{n_i-1}[B_{Ca,i}]}{K_{Ca,i}^{n_i}\left(1+[Ca^{2+}]^{n_i}\ /\ K_{Ca,i}^{n_i}\right)^2} \tag{2.36x}$$

Table S2.2.1. Additional Mitochondrial Buffering Parameters

Parameter	Definition	Value	Reference
$[B_H]$	Total binding sites	1.8 M	[a]
K_H	H^+ binding constant	$10^{-7.5}$ M	[a]
$[B_{Mg}]$	Total binding sites	13 mM	[10][b]
K_{Mg}	Mg^{2+} binding constant	340 µM	[10][b]
$[B_{Ca,1}]$	Class 1 total binding sites	35.0 mM	[a]
$K_{Ca,1}$	Class 1 Ca^{2+} binding constant	2.0 µM	[a]
n_1	Class 1 number of Ca^{2+} binding sites	1 unitless	[a]
$[B_{Ca,2}]$	Class 2 total binding sites	15.0 mM	[a]
$K_{Ca,2}$	Class 2 Ca^{2+} binding constant	1.7 µM	[a]
n_2	Class 2 number of Ca^{2+} binding sites	6 unitless	[a]

[a] Fit by simulating the experimental data sets described in the main paper
[b] Fit to data in [10].

Additional Buffering in the Extra-Mitochondrial Compartment

Both MOPS and EGTA were included in the incubation medium for the conditions simulated in the main paper. Each of these reagents buffers H^+s, and EGTA also buffers Ca^{2+}. Moreover, EGTA is primarily dibasic in the pH range simulated. To account for these additional buffers in the extra-mitochondrial compartment, Eqs. 2.16, 2.20, 2.36 and 2.40 are modified by adding the following terms:

$$H_{\partial H} = H_{\partial H} + \frac{[MOPS]}{K_{MOPS}^{H}\left(1+[H^+]/K_{MOPS}^{H}\right)^2} + \frac{[EGTA]\left(2[H^+]+K_{HEGTA}^{H}\right)\left([Ca^{2+}]+K_{EGTA}^{Ca}\right)}{K_{EGTA}^{Ca}K_{EGTA}^{H}K_{HEGTA}^{H}P_{EGTA}^{2}} \tag{2.16e}$$

$$H_{\partial Ca} = H_{\partial Ca} - \frac{[EGTA][H^+]\left([H^+]+K_{HEGTA}^{H}\right)}{K_{EGTA}^{Ca}K_{EGTA}^{H}K_{HEGTA}^{H}P_{EGTA}^{2}} \tag{2.20e}$$

$$Ca_{\partial H} = Ca_{\partial H} - \frac{[EGTA][Ca^{2+}]\left(2[H^+]+K_{HEGTA}^{H}\right)}{K_{EGTA}^{Ca}K_{EGTA}^{H}K_{HEGTA}^{H}P_{EGTA}^{2}} \tag{2.36e}$$

$$Ca_{\partial Ca} = Ca_{\partial Ca} + \frac{[EGTA]\left([H^+]^2+[H^+]K_{HEGTA}^{H}+K_{EGTA}^{H}K_{HEGTA}^{H}\right)}{K_{EGTA}^{Ca}K_{EGTA}^{H}K_{HEGTA}^{H}P_{EGTA}^{2}} \qquad (2.36e)$$

where the binding polynomial for EGTA is

$$P_{EGTA} = 1+[H^+]/K_{EGTA}^{H}+[H^+]^2/K_{HEGTA}^{H}/K_{EGTA}^{H}+[Ca^{2+}]/K_{EGTA}^{Ca} \tag{2.42}$$

Flux Terms

The terms representing flux into a given compartment are defined as

$$\Phi_H = -\sum_{i=1}^{Nr}\frac{\partial[H_{bound}]}{\partial[L_i]}\frac{d[L_i]}{dt}+\sum_{k=1}^{Nk}n_k J_k + J_t^H \tag{2.43}$$

$$\Phi_{Na} = -\sum_{i=1}^{Nr}\frac{\partial[Na_{bound}]}{\partial[L_i]}\frac{d[L_i]}{dt}+J_t^{Na} \tag{2.44}$$

$$\Phi_K = -\sum_{i=1}^{Nr}\frac{\partial[K_{bound}]}{\partial[L_i]}\frac{d[L_i]}{dt}+J_t^{K} \tag{2.45}$$

$$\Phi_{Mg} = -\sum_{i=1}^{Nr}\frac{\partial[Mg_{bound}]}{\partial[L_i]}\frac{d[L_i]}{dt}+J_t^{Mg} \tag{2.46}$$

and

$$\Phi_{Ca} = -\sum_{i=1}^{Nr} \frac{\partial[Ca_{bound}]}{\partial[L_i]} \frac{d[L_i]}{dt} + J_t^{Ca} \tag{2.47}$$

where N_k is the number of reactions, n_k is the stoichiometric coefficient of k^{th} reaction, J_k is the k^{th} reaction rate and J_t^i is the i^{th} cation transport rate into the compartment.

The ligand dependent partial derivative expressions are defined as

$$\frac{\partial[H_{bound}]}{\partial[L_i]} = [H^+]\sum_{i=1}^{Nr} \frac{1}{K_i^H P_i} \tag{2.48}$$

$$\frac{\partial[Na_{bound}]}{\partial[L_i]} = [Na^+]\sum_{i=1}^{Nr} \frac{1}{K_i^{Na} P_i} \tag{2.49}$$

$$\frac{\partial[K_{bound}]}{\partial[L_i]} = [K^+]\sum_{i=1}^{Nr} \frac{1}{K_i^K P_i} \tag{2.50}$$

$$\frac{\partial[Mg_{bound}]}{\partial[L_i]} = [Mg^{2+}]\sum_{i=1}^{Nr} \frac{1}{K_i^{Mg} P_i} \tag{2.51}$$

and

$$\frac{\partial[Ca_{bound}]}{\partial[L_i]} = [Ca^{2+}]\sum_{i=1}^{Nr} \frac{1}{K_i^{Ca} P_i} . \tag{2.52}$$

Compartment Reactions and Fluxes

The generation of protons by biochemical reactions in the mitochondrial is defined as

$$\sum_{k=1}^{Nk} n_k J_k = J_{DH} . \tag{2.53}$$

This term is zero in the extra-mitochondrial compartment.

The transport of cations into the mitochondrial compartment are defined as

$$J_t^H = \left(\left(nH_{F_1F_O} - 1\right) J_{F_1F_O} + 2J_{PIC} + J_{KHE} + J_{NHE} + 2J_{CHE} + J_{Hleak} - 5J_{C1} - 2J_{C3} - 4J_{C4}\right) \tag{2.54}$$

$$J_t^{Na} = \left(3J_{NCE} - J_{NHE}\right) \tag{2.55}$$

$$J_t^{K} = -J_{KHE} \tag{2.56}$$

$$J_t^{Mg} = 0 \tag{2.57}$$

and

$$J_t^{Ca} = \left(2J_{CU} + J_{RaM} - J_{NCE} - J_{CHE}\right). \tag{2.58}$$

The transport terms for the extra-mitochondrial compartment are the negative of their mitochondrial counterparts.

S2.3: Model Rate Equations

Each reaction rate and transport mechanism in the model is presented below. First, the electron transport related reaction rates are shown. Next, the oxidative phosphorylation related reaction rates and transport mechanisms are discussed. Finally, the cation transport mechanisms are presented.

S2.3A - Electron Transport Related Reaction Rates

Substrate oxidation and the electron transport system are based on the rate equations presented in Beard et al. [1] and Wu et al. [2]. The mitochondrial dehydrogenase rate (substrate oxidation rate) will be introduced first, followed by the electron transport system rate equations and finally the passive proton leak rate equation is presented.

Mitochondrial Dehydrogenases

The biochemical equation for the Mitochondrial dehydrogenase is defined as

$$NAD_m^+ \rightleftarrows NADH_m + H_m^+.$$

The rate expression used in the model is

$$J_{\mathrm{DH}} = X_{\mathrm{DH}} \left(\frac{1+[\mathrm{Pi}]_m / k_{\mathrm{Pi},1}}{1+[\mathrm{Pi}]_m / k_{\mathrm{Pi},2}} \right) \left(r_{\mathrm{DH}}\ [\mathrm{NAD}^+]_m - [\mathrm{NADH}]_m \right).$$

Table S2.3.1. Mitochondrial Dehydrogenase Parameters

Parameter	Definition	Value	Reference
X_{DH}	Dehydrogenase rate	20200 nmol/min/mg	[1]
r_{DH}	Steady state NADH/NAD ratio	4.58 unitless	[1]
$k_{Pi,1}$	1st Phosphate activation constant	0.134 mM	[1]
$k_{Pi,2}$	2nd Phosphate activation constant	0.677 mM	[1]

NADH-ubiquinone reductase: Complex I

The biochemical equation for Complex I is defined as

$$UQ_m + NADH_m + 5H_m^+ \rightleftarrows NAD_m^+ + UQH_{2,m} + 4H_e^+.$$

The equilibrium constant is defined as

$$K_{eq} = e^{-\left(\Delta_r G_{CI}^0 - 4\Delta\Psi\right)/RT} \left(\left[H^+\right]_m^5 / \left[H^+\right]_e^4 \right).$$

The rate expression used in the model is

$$J_{CI} = X_{C1} \left(K_{eq}[NADH]_m[UQ]_m - [NAD]_m[UQH_2]_m \right).$$

Table S2.3.2. NADH-Ubiquinone Reductase Parameters

Parameter	Definition	Value	Reference
X_{CI}	Complex I activity	$7.2\text{x}10^9$ nmol/min/mg/M^2	[2]
$\Delta_r G_{CI}^0$	Gibb's free energy of reaction	-109.7 kJ/mol	[2]

Ubiquinol-cytochrome-c reductase: Complex III

The biochemical equation for Complex III is defined as

$$UQH_{2,m} + 2CytC_i^{3+} + 2H_m^+ \rightleftarrows UQ_m + 2CytC_i^{2+} + 4H_e^+ .$$

The equilibrium constant is defined as

$$K_{eq} = e^{-\left(\Delta_r G_{C3}^0 - 2\Delta\Psi\right)/RT} \left(\left[H^+\right]_m^2 / \left[H^+\right]_e^4\right).$$

The rate expression used in the model is

$$J_{C3} = X_{C3} \left(\frac{1 + [Pi^{2-}]_m / k_{Pi,3}}{1 + [Pi^{2-}]_m / k_{Pi,4}} \right) \left(K_{eq}^{1/2} [UQH_2]_m^{1/2} [CytC^{3+}]_i - [UQ]_m^{1/2} [CytC^{2+}]_i \right).$$

Table S2.3.3. Ubiquinol-Cytochrome-c Reductase Parameters

Parameter	Definition	Value	Reference
X_{C3}	Complex III activity	1.74x10^5 nmol/min/mg/M$^{3/2}$	[2]
$\Delta_r G_{C3}^0$	Gibb's free energy of reaction	55.46 kJ/mol	[2]
K_{Pi}	Pi binding constant	32.5 µM	[2]
β_{Pi}	Pi activation constant	3.5 mM	[2]

Cytochrome-c oxidase: Complex IV

The biochemical equation for Complex III is defined as

$$2CytC_i^{2+} + \frac{1}{2}O_{2,m} + 4H_e^+ \rightleftarrows 2CytC_i^{3+} + H_2O + 2H_e^+ .$$

The equilibrium constant is defined as

$$K_{eq} = e^{-\left(\Delta_r G_{C4}^0 - 4\Delta\Psi\right)/RT} \left(\left[H^+\right]_m^4 / \left[H^+\right]_i^2 \right).$$

The rate expression used in the model is

$$J_{C4} = X_{C4} \left(\frac{e^{\frac{F\Delta\Psi}{RT}}}{1 + K_{O_2} / [O_2]_m} \right) \left(\frac{[CytC^{2+}]_i}{[CytC^{2+}]_i + [CytC^{3+}]_i} \right) \left(K_{eq}^{1/2} [CytC^{2+}]_i [O_2]_m^{1/4} - [CytC^{3+}]_i \right).$$

Table S2.3.4. Cytochrome-c Oxidase Parameters

Parameter	Definition	Value	Reference
X_{C4}	Complex IV activity	24 nmol/min/mg/M	[2]
$\Delta_r G_{C4}^0$	Gibb's free energy of reaction	-202.16 kJ/mol	[2]
K_{O_2}	O_2 binding constant	$120\text{x}10^{-6}$ M	[2]

Proton leak

The electrophoretically driven proton uptake via leak pathways was updated to account for the highly non-linear nature of the leak across energy-transducing membranes [11]. The proton leak permeability, P_{Hleak}, is set to $8.06\text{x}10^7$ nmol/min/mg/M so as to simulate the same state 2 respiration results in [2].

$$J_{Hleak} = P_{Hleak} \left([H^+]_e e^{\left(\frac{F\Delta\Psi}{2RT}\right)} - [H^+]_m e^{-\left(\frac{F\Delta\Psi}{2RT}\right)} \right)$$

S2.3B - Oxidative Phosphorylation Related Reactions Rates

Adenine nucleotide translocase

The biochemical equation for ANT is defined as

$$ATP_m + ADP_e \rightleftarrows ATP_e + ADP_m$$

The rate expression used in the model is

$$J_{\mathrm{ANT}} = \left(\frac{X_{\mathrm{ANT}}}{1 + k_{\mathrm{ADP}} / [\mathrm{ADP}^{3-}]_e}\right)\left(\frac{[\mathrm{ADP}^{3-}]_e}{[\mathrm{ADP}^{3-}]_e + [\mathrm{ATP}^{4-}]_e e^{(\theta-1)F\Delta\Psi/RT}} - \frac{[\mathrm{ADP}^{3-}]_m}{[\mathrm{ADP}^{3-}]_m + [\mathrm{ATP}^{4-}]_m e^{\theta F\Delta\Psi/RT}}\right).$$

Table S2.3.5. Adenine Nucleotide Translocase Parameters

Parameter	Definition	Value	Reference
X_{ANT}	ANT activity	1.66 nmol/min/mg	[2]
k_{ADP}	ADP binding constant	3.5 µM	[2]
θ	Fractional position of energy barrier peak	0.65	[2]

F_1F_O ATP synthase

The biochemical equation for F_1F_O ATP synthase is defined as

$$ADP_m + Pi_m \rightleftarrows ATP_m \, .$$

The equilibrium constant is defined as

$$K_{eq} = e^{-\left(\Delta_r G^0_{F_1F_O} + 3F\Delta\Psi/RT\right)}\left(\frac{[H^+]_e^{nH}}{[H^+]_m^{nH-1}}\right)\left(\frac{P_{ATP}}{P_{ADP}P_{Pi}}\right).$$

The rate expression used in the model is

$$J_{F_1F_O} = X_{F_1F_O}\left([ADP]_m [Pi]_m K_{eq} - [ATP]_m\right).$$

Table S2.3.6. F_1F_O ATP synthase Parameters

Parameter	Definition	Value	Reference
V_{F1FO}	F_1F_O activity	$1.8\text{x}10^7$ nmol/min/mg/M	[2][a]
$\Delta_r G^0_{F_1F_O}$	Gibb’s free energy of reaction	-4.51 kJ/mol	[2]
nH	H^+:ATP ratio	3	[2]

[a] The activity was lowered from the cited value to reduced the stiffness of the system of ODEs. This resulted in negligible differences in state variable dynamics for the simulations that used to identify this parameter value.

Inorganic phosphate carrier

The biochemical equation for the PIC is defined as

$$Pi_e + H_e^+ \rightleftarrows Pi_m + H_m^+ .$$

The rate expression used in the model is

$$J_{\mathrm{PIC}} = X_{\mathrm{PIC}} \left(\frac{[Pi^-]_m[\mathrm{H}^+]_m - [Pi^-]_e[\mathrm{H}^+]_e}{[Pi^-]_e + k_{\mathrm{PIC}}} \right).$$

Table S2.3.7 Inorganic Phosphate Parameters

Parameter	Definition	Value	Reference
X_{PIC}	PIC activity	7.35x10^{12} nmol/min/mg/M	[2]
k_{PIC}	Pi binding constant	1.6 mM	[2]

S2.3C - Cation Related Reaction Rates

Potassium-hydrogen exchanger

The biochemical equation for KHE is defined as

$$K_m^+ + H_e^+ \rightleftarrows K_e^+ + H_m^+ .$$

The rate expression used in the model is

$$J_{KHE} = X_{KHE} \left([K^+]_m[H^+]_e - [K^+]_e[H^+]_m \right)$$

where X_{KHE} is set to 4.3x10^{11} nmol/min/mg/M^2 as in [2].

Calcium-hydrogen exchanger

The biochemical equation for CHE is defined as

$$Ca_m^{2+} + 2H_e^+ \rightleftarrows Ca_e^{2+} + 2H_m^+ .$$

The rate expression used in the model is

$$J_{CHE} = X_{CHE}\left([Ca^{2+}]_m[H^+]_e^2 - [Ca^{2+}]_e[H^+]_m^2\right)$$

where X_{CHE} is set to 4.0x10^{19} nmol/min/mg/M^3. This value was fit by simulating the experimental data sets described in the main paper.

Sodium-hydrogen exchanger

The biochemical equation for NHE is defined as

$$Na_m^+ + H_e^+ \rightleftarrows Na_e^+ + H_m^+ .$$

The rate expression used in the model is

$$J_{\mathrm{NHE}} = \frac{X_{\mathrm{NHE}}[\mathrm{H}^+]_m^3}{\left(K_{iH}^3 + [\mathrm{H}^+]_m^3\right)} \times \left(\frac{[\mathrm{H}^+]_e[\mathrm{Na}^+]_m - [\mathrm{H}^+]_m[\mathrm{Na}^+]_e}{K_{Na}\left([\mathrm{H}^+]_e + [\mathrm{H}^+]_m\right) + [\mathrm{H}^+]_e^2[\mathrm{Na}^+]_m + [\mathrm{H}^+]_m[\mathrm{Na}^+]_e} \right).$$

Table S2.3.8. Sodium-Hydrogen Exchanger Parameters [a]

Parameter	Definition	Value	Reference
X_{NHE}	NHE activity	720 nmol/min/mg	[1]
K_{iH}	H^+ inactivation constant	$10^{-7.2}$ M	[1]
K_{Na}	Na^+ binding constant	25 mM	[1]

[a] The original hill coefficient for the H^+ regulatory mechanism was 2. Increasing it to 3 produces better fits to the data in [12].

Rapid Mode of Calcium Uptake

The biochemical equation for RaM is defined as

$$Ca_e^{2+} \rightleftarrows Ca_m^{2+}.$$

The rate expression used in the model is

$$J_{RaM} = X_{RaM} O \left(\frac{[Ca^{2+}]_e / K_{Ca} e^{\Delta\Phi/2} - [Ca^{2+}]_m / K_{Ca} e^{-\Delta\Phi/2}}{1 + [Ca^{2+}]_e / K_{Ca} + [Ca^{2+}]_m / K_{Ca}} \right),$$

where the fraction of RaM in the O state is

$$O = f_{Ca} RO,$$

the relationship between the rest and open states is defined as

$$f_{Ca} = \frac{[Ca^{2+}]_e^n}{[Ca^{2+}]_e^n + K_O^n},$$

and the relationship between the 1st and 2nd inhibited states is defined as

$$g_{Ca} = \frac{[Ca^{2+}]_e^n}{[Ca^{2+}]_e^n + K_I^n}.$$

Table S2.3.9 Rapid Mode of Calcium Uptake Parameters

Parameter	Definition	Value	Reference
X_{RaM}	RaM activity	0.25 nmol/min/mg	[3][a]
K_{Ca}	RaM Complex Ca^{2+} affinity	340 nM	[3]
n	Ca^{2+} cooperativity	24 unitless	[3]
K_O	State *O* Ca^{2+} binding constant	224 nM	[3]
K_I	State I_1 Ca^{2+} binding constant	110 nM	[3]
k_{OI_1}	State O-I_1 transition rate	100 s^{-1}	[3]
k_{I_1O}	State I_1-O transition rate	3.71x10^{-4} s^{-1}	[3][b]
k_{I_2R}	State I_2-R transition rate	0.0384 s^{-1}	[3]
k_{RI_2}	State R-I_2 transition rate	4x10^{-4} s^{-1}	[3][c]

[a] Fit by simulating the experimental data sets described in the main paper.

[b] Thermodynamically constrained via microscopic reversibility where $k_{I_1O} = k_{OI_1} k_{I_2R} K_I^n / k_{RI_2} / K_O^n$.

[c] Increased by 100 from the cited value to remove the "Ca^{2+} leak" phenomenon that occurs when $[Ca^{2+}]_e$ is near K_O. See [3] for details.

Calcium uniporter

The biochemical equation for CU is defined as

$$2Ca_e^{2+} \rightleftarrows 2Ca_m^{2+}.$$

The rate expression used in the model is

$$J_{\mathrm{CU}} = X_{CU}\left(k_i \frac{[\mathrm{Ca}^{2+}]_\mathrm{e}^2}{K_\mathrm{Ce}^2} - k_o \frac{[\mathrm{Ca}^{2+}]_m^2}{K_\mathrm{Cx}^2}\right) / D$$

where the binding constants are

$$\begin{aligned}
K_\mathrm{Ce} &= K_{Ca}(1+[\mathrm{Pi}]_\mathrm{e}/(K_\mathrm{Pi}+[\mathrm{Pi}]_\mathrm{e}),\\
K_\mathrm{Cm} &= K_{Ca}(1+[\mathrm{Pi}]_m/(K_\mathrm{Pi}+[\mathrm{Pi}]_m),\\
K_\mathrm{Me} &= K_{Mg}\ /\ (1+[\mathrm{Pi}]_\mathrm{e}/(K_\mathrm{Pi}+[\mathrm{Pi}]_\mathrm{e}),\\
K_\mathrm{Mm} &= K_{Mg}\ /\ (1+[\mathrm{Pi}]_m/(K_\mathrm{Pi}+[\mathrm{Pi}]_m),
\end{aligned}$$

and the $\Delta\Psi$-dependent translocation terms are

$$\begin{aligned}
k_\mathrm{i} &= \mathrm{e}^{\left(\Delta\Phi + nH\ln\left\{\left(\Delta\Phi/nH\right)/\sinh\left(\Delta\Phi/nH\right)\right\}\right)},\\
k_\mathrm{o} &= e^{\left(\Delta\Phi - nH\ln\left\{\left(\Delta\Phi/nH\right)/\sinh\left(\Delta\Phi/nH\right)\right\}\right)},
\end{aligned}$$

with the denominator defined as

$$\begin{aligned}
D=1&+\frac{[\mathrm{Ca}^{2+}]_\mathrm{e}}{K_\mathrm{Ce}}+\frac{[\mathrm{Ca}^{2+}]_\mathrm{e}^2}{K_\mathrm{Ce}^2}+\frac{[\mathrm{Ca}^{2+}]_m}{K_\mathrm{Cm}}+\frac{[\mathrm{Ca}^{2+}]_m^2}{K_\mathrm{Cm}^2}+\frac{[\mathrm{Mg}^{2+}]_\mathrm{e}}{K_\mathrm{Me}}+\frac{[\mathrm{Mg}^{2+}]_\mathrm{e}^2}{K_\mathrm{Me}^2}+\frac{[\mathrm{Mg}^{2+}]_m}{K_\mathrm{Mm}}+\frac{[\mathrm{Mg}^{2+}]_m^2}{K_\mathrm{Mm}^2}\\
&+\frac{[\mathrm{Ca}^{2+}]_\mathrm{e}[\mathrm{Mg}^{2+}]_\mathrm{e}}{\gamma\cdot K_\mathrm{Ce}\cdot K_\mathrm{Me}}+\frac{[\mathrm{Ca}^{2+}]_\mathrm{e}[\mathrm{Mg}^{2+}]_\mathrm{e}^2}{\gamma^2\cdot K_\mathrm{Ce}\cdot K_\mathrm{Me}^2}+\frac{[\mathrm{Ca}^{2+}]_\mathrm{e}^2[\mathrm{Mg}^{2+}]_\mathrm{e}}{\gamma^2\cdot K_\mathrm{Ce}^2\cdot K_\mathrm{Me}}+\frac{[\mathrm{Ca}^{2+}]_\mathrm{e}^2[\mathrm{Mg}^{2+}]_\mathrm{e}^2}{\gamma^4\cdot K_\mathrm{Ce}^2\cdot K_\mathrm{Me}^2}+\frac{[\mathrm{Ca}^{2+}]_m[\mathrm{Mg}^{2+}]_m}{\gamma\cdot K_\mathrm{Cm}\cdot K_\mathrm{Mm}}\\
&+\frac{[\mathrm{Ca}^{2+}]_m[\mathrm{Mg}^{2+}]_m^2}{\gamma^2\cdot K_\mathrm{Cm}\cdot K_\mathrm{Mm}^2}+\frac{[\mathrm{Ca}^{2+}]_m^2[\mathrm{Mg}^{2+}]_m}{\gamma^2\cdot K_\mathrm{Cm}^2\cdot K_\mathrm{Mm}}+\frac{[\mathrm{Ca}^{2+}]_m^2[\mathrm{Mg}^{2+}]_m^2}{\gamma^4\cdot K_\mathrm{Cm}^2\cdot K_\mathrm{Mm}^2}
\end{aligned}$$

and $\Delta\Phi = F\Delta\Psi / RT$.

Table S2.3.10. Calcium Uniporter Parameters

Parameter	Definition	Value	Reference
X_{CU}	CU activity	1.5 nmol/min/mg	[5][a]
nH	Hill coefficient for ΔΨ dependence	2.65 unitless	[5]
K_{Ca}	Ca^{2+} binding constant	3.64 μM	[5][a]
K_{Mg}	Mg^{2+} binding constant	550 μM	[5]
K_{Pi}	Pi inhibition constant	200 μM	[5]
γ	Mg^{2+} induced binding modulation	5.7 unitless	[5]

[a] Fit by simulating the experimental data sets described in the main paper.

Sodium-calcium exchanger

The biochemical equation for NCE is defined as

$$Ca_m^{2+} + 3Na_e^{+} \rightleftarrows Ca_e^{2+} + 3Na_m^{+}.$$

The rate expression used in the model is

$$J_{\mathrm{NCE}} = \frac{\mathrm{X_{NCE}}}{D}\left(k_a . D_{\mathrm{Hm}} \frac{[\mathrm{Ca}^{2+}]_m [\mathrm{Na}^+]_{\mathrm{e}}^n}{K'_{\mathrm{Cx}} . K_{Na}^n} - k_b . D_{\mathrm{He}} \frac{[\mathrm{Ca}^{2+}]_{\mathrm{e}} [\mathrm{Na}^+]_m^n}{K'_{\mathrm{Ce}} . K_{Na}^n} \right)$$

where the binding constants are

$$K'_{\mathrm{Cm}} = K_{Ca}\left(\sum_{q=1}^{m} \frac{[\mathrm{H}^+]_m^q}{K_{\mathrm{H1}}^q} + 1 + \sum_{q=1}^{m} \frac{K_{\mathrm{H2}}^q}{[\mathrm{H}^+]_m^q} \right)$$

$$K'_{\mathrm{Ce}} = K_{Ca}\left(\sum_{q=1}^{m} \frac{[\mathrm{H}^+]_{\mathrm{e}}^q}{K_{\mathrm{H1}}^q} + 1 + \sum_{q=1}^{m} \frac{K_{\mathrm{H2}}^q}{[\mathrm{H}^+]_{\mathrm{e}}^q} \right)$$

and the ΔΨ-dependent translocation terms are

$$k_a = \mathrm{e}^{[(n\beta z_{\mathrm{Na}} - \beta z_{\mathrm{Ca}})\Delta\Phi]},$$

$$k_b = \mathrm{e}^{[(-n\beta z_{\mathrm{Na}} + \beta z_{\mathrm{Ca}})\Delta\Phi]}$$

with the denominator terms defined as

$$D_{\mathrm{Hm}} = \left(\sum_{q=1}^{m} \frac{[\mathrm{H}^+]_m^q}{K_{\mathrm{H1}}^q} + 1 + \sum_{q=1}^{m} \frac{K_{\mathrm{H2}}^q}{[\mathrm{H}^+]_m^q} \right)\left(\frac{[\mathrm{H}^+]_m^m}{K_{\mathrm{H2}}^m} \right)$$

$$D_{\mathrm{He}} = \left(\sum_{q=1}^{m} \frac{[\mathrm{H}^+]_{\mathrm{e}}^q}{K_{\mathrm{H1}}^q} + 1 + \sum_{q=1}^{m} \frac{K_{\mathrm{H2}}^q}{[\mathrm{H}^+]_{\mathrm{e}}^q} \right)\left(\frac{[\mathrm{H}^+]_{\mathrm{e}}^m}{K_{\mathrm{H2}}^m} \right)$$

$$D = D_{\mathrm{Hm}}\left(1 + \sum_{p=1}^{n} \frac{[\mathrm{Na}^+]_{\mathrm{e}}^p}{K'^{\,p}_{\mathrm{Ne}}} + \frac{[\mathrm{Ca}^{2+}]_m}{K'_{\mathrm{Cm}}} + \sum_{p=1}^{n} \frac{[\mathrm{Ca}^{2+}]_m [\mathrm{Na}^+]_{\mathrm{e}}^p}{K'_{\mathrm{Cm}} . K'^{\,p}_{\mathrm{Ne}}} \right) + D_{\mathrm{He}}\left(1 + \sum_{p=1}^{n} \frac{[\mathrm{Na}^+]_m^p}{K'^{\,p}_{\mathrm{Nm}}} + \frac{[\mathrm{Ca}^{2+}]_{\mathrm{e}}}{K'_{\mathrm{Ce}}} + \sum_{p=1}^{n} \frac{[\mathrm{Ca}^{2+}]_{\mathrm{e}} [\mathrm{Na}^+]_m^p}{K'_{\mathrm{Ce}} . K'^{\,p}_{\mathrm{Nm}}} \right) - 1$$

and $\Delta\Phi = F\Delta\Psi / RT$.

Table S2.3.11. Sodium-Calcium Exchanger Parameters

Parameter	Definition	Value	Reference
X_{NCE}	NCE activity	0.5056 nmol/min/mg	[4][a]
β	$\Delta\Psi$-related Energy barrier parameter	0.5 unitless	[4]
K_{Na}	Sodium binding constant	2.0 mM	[4][a]
K_{Ca}	Calcium binding constant	110 nM	[4][a]
K_{H1}	1st Proton binding constant	$10^{-7.19}$ M	[4]
K_{H2}	2nd Proton binding constant	$10^{-6.85}$ M	[4]
n	Sodium:calcium stoichiometry	3	[4]
m	Proton regulatory parameter	3	[4][a]

[a] Fit by simulating the experimental data sets described in the main paper.

The proton regulatory parameter of the NCE was originally estimated as 6 [4]; however, a better fit to the data presented in the main paper is obtained when this parameter is adjusted to 3. Figure S2 demonstrates the tradeoff in fits when changing this parameter. Figures S2A and B are the model predictions compared to the data for the original parameter set identified in [4]. The model fits the $[Ca^{2+}]_e$ dependence at different pH's well, but it cannot fit the pH dependence in the physiological range. Figures S2C and D shows that when the proton dependency is adjusted to 3, the opposite is observed. Moreover, the $[Ca^{2+}]_e$ dependence trend in still maintained as shown in Figure S2C. To fit the pH dependency, parameter K_{Ca} is changed to 110 nM versus the original value of 2.26 nM. The inability of the model to fit both data sets simultaneously is attributed to inconsistencies in the data set itself (see [4] for discussion).

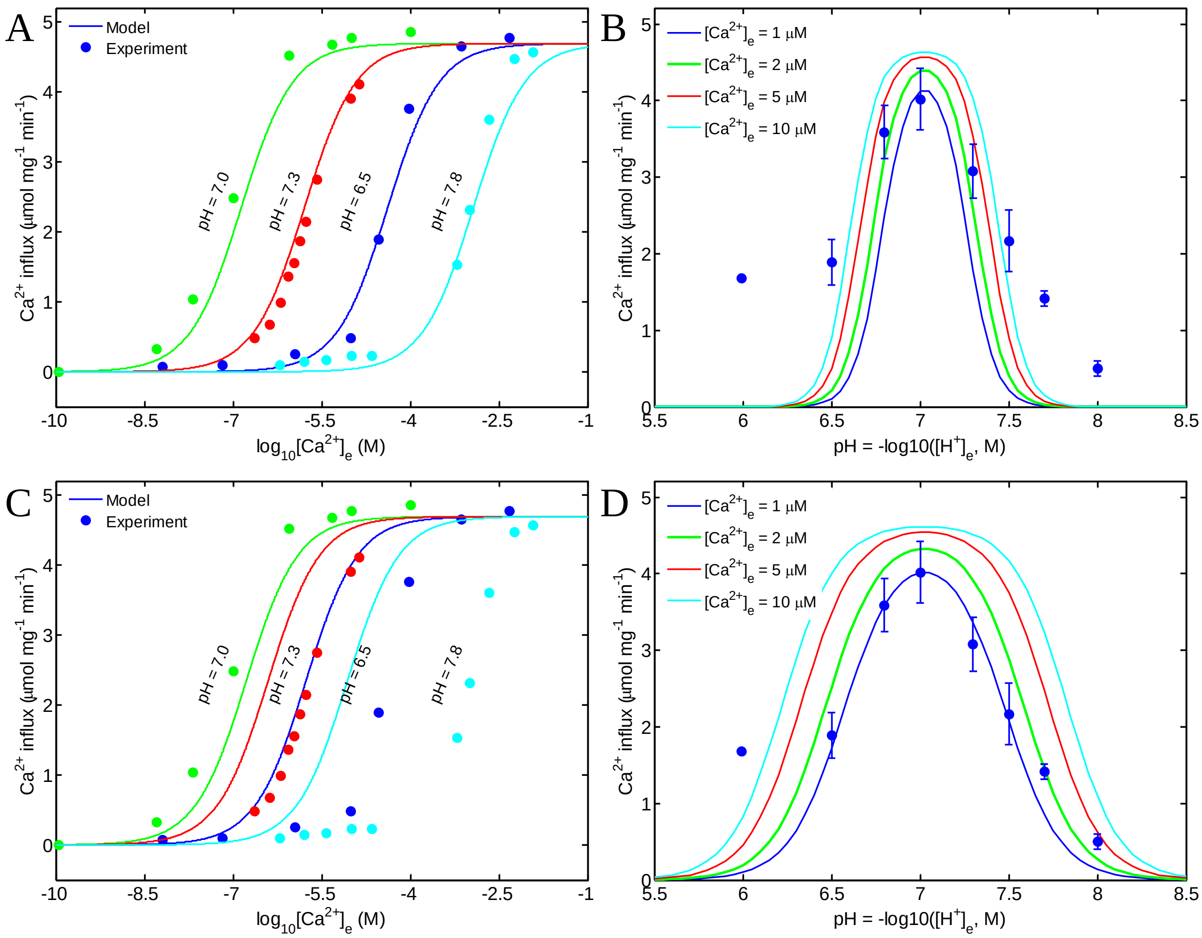

Figure S2. Changing the proton regulatory parameter allows the model to fit either the $[Ca^{2+}]_e$ dependence or pH dependence, not both. The model predictions compared to the data for the original parameter set is shown in A and B for the $[Ca^{2+}]_e$ dependence and pH dependence, respectively. Changing the proton regulatory parameter to 3 and the Ca^{2+} binding constant to 11.3 nM maintains the same trend in the $[Ca^{2+}]_e$ dependence data and facilitates fits to the pH dependence data as shown in C and D, respectively.

Supplement 3: A Proton Regulatory Parameter of 6 for the NCE Fails to Predict the Data

The best fit to the data presented in the main paper when the proton dependency was kept at its original value of 6 is shown in Figure S3. For the $CaCl_2$ boluses up to 20 µM, the model fits the data equally well to the optimal parameter set. For boluses larger than 20 µM, the model fails to predict the Ca^{2+} extrusion rate as described by the data. This is because of the alkalization of the mitochondrial matrix upon Ca^{2+} uptake causing a reduction in Ca^{2+} affinity for the NCE. Thus, the proton regulatory parameter was reduced to 3 in order to simulate the level of Ca^{2+} extrusion for the 30 and 40 µM $CaCl_2$ boluses.

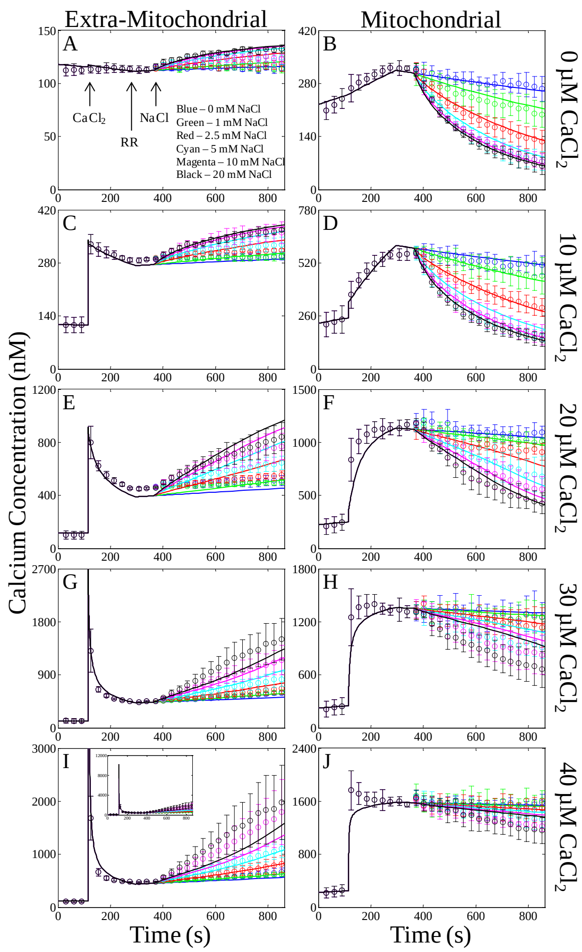

Figure S3. Optimal fits to the Ca^{2+} uptake, sequestration and release data when the proton regulatory parameter of the NCE is fixed at 6. The simulation results of each bolus $CaCl_2$ addition are plotted by row with the extra-mitochondrial (left) and mitochondrial (right) Ca^{2+} concentrations plotted by column. The color of each lines corresponds to the following: blue, 0 mM; green 1 mM; red, 2.5 mM; cyan, 5 mM; magenta, 10 mM; and black, 20 mM NaCl bolus. The times that reagents were added to the mitochondria are indicated by the arrows. When the $CaCl_2$ bolus exceeds 20 µM, the model is unable to reproduce the Ca^{2+} extrusion rates described by the data.

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